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ADJUSTABLE HOSE CLAMPING ARRANGEMENT

Abstract

An adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold. The adjustable hose clamping arrangement includes a clamping device configured for fixedly clamping the hose such that a relative movement between the hose and the clamping device is inhibited, a mold bracket configured for being mountable to the steel branch of the vacuum mold, and two connecting devices, wherein a first end of each of the two connecting devices is connected to the clamping device and a second end of each of the two connecting devices being opposite of the first end, is connected to the mold bracket such that an outer surface of the hose next to a free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a national stage of PCT Application No. PCT/EP2023/059664, having a filing date of Apr. 13, 2023, which claims priority to EP Application No. 22173198.7, having a filing date of May 13, 2022, the entire contents both of which are hereby incorporated by reference.

FIELD OF TECHNOLOGY

[0002] The following relates to the field of vacuum assisted resin transfer molding. Particularly, the following relates to an adjustable hose clamping arrangement, a mold arrangement comprising an adjustable hose clamping arrangement, a method of manufacturing an adjustable hose clamping arrangement, and a method of using an adjustable hose clamping arrangement.

BACKGROUND

[0003] In today's vacuum assisted resin transfer molding process, an outer rigid mold shell and an inner flexible mold shell are used. Hose lead-ins are used for supplying resin to the mold cavity holding the dry lay-up and vacuum for internal membrane filters. These lead-ins are in the form of steel branches with a certain degree of distance between the inner surface of the branch and the outer surface of the hose or molded plastic insert used.

[0004] On the outside of the mold a rubber cold shrink tube is used to seal the hose to the branch, in combination with either tacky tape and/or O-rings. The cold shrink tube is a highly flexible tube in black ethylene propylene diene monomer (EPDM) rubber which after the unwinding of a flexible core allows the pre-stretched rubber to cling to the branch and to form a seal. However, once mounted, the cold shrink tube completely hides the parts in need for a perfect vacuum-tight seal. Furthermore, the cold shrink tube is a rather expensive component not originally intended for a one-time use. The cold shrink tube has rather been designed for primary insulation of solid dielectric insulated wires and cable splicing rated to 1000 volts. Additionally, when removing the cold shrink tube, the cold shrink tube has to be split by a knife. This over time ruins the steel surface of the steel branch. This may result in leakages which occur and may not be realized.

[0005] Furthermore, hiding the parts in need for a perfect vacuum-tight seal may result in less chance of realising a possible fault and leaking throughout the casting process adding to the costs of later repair and error codes related to air intruding into the mold.

[0006] Therefore, it may be a need to provide a device for allowing to reliably seal, in particular vacuum-tight seal, a mold in a vacuum assisted resin transfer molding process which is cost-saving and reusable.

SUMMARY

[0007] An aspect relates to a device for allowing to seal a mold in a reliable and cost-saving way. Additionally, it may be desirable to enable an inspection of the sealing, particularly vacuum-tight sealing, of a vacuum mold during the molding process. Hence, an aspect relates to an objective of the present invention to provide an adjustable hose clamping arrangement for a vacuum mold with an increased service life, reduced operational costs, and allowing an increased sealing, particularly vacuum-tight sealing, of the vacuum mold.

[0008] This aspect may be solved by the adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, a vacuum mold arrangement comprising an adjustable hose clamping arrangement and a vacuum mold, a method of manufacturing an adjustable hose clamping arrangement, and a method of using an adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold.

[0009] According to an aspect of embodiments of the present invention, an adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, wherein the flexible insert is guidable through the steel branch from an inside of the vacuum mold to an outside of the vacuum mold, is described. The adjustable hose clamping arrangement comprises a clamping device configured for fixedly clamping the hose such that a relative movement between the hose and the clamping device is inhibited, a mold bracket configured for being mountable to the steel branch of the vacuum mold, two connecting devices, wherein a first end of each of the two connecting devices is connected to the clamping device and a second end of each of the two connecting devices being opposite of the first end, is connected to the mold bracket such that an outer surface of the hose next to a free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.

[0010] Embodiments of the present invention are based on the idea that the adjustable hose clamping arrangement allows for a vacuum-tight sealing by a tape, in particular butyl tape, and does not need any additional sealing material such as for example a cold shrink tube. The sealing allows for continuous visual inspection and easy mending in case a leaking occurs. As the sealing is directly visible for an operator, an ultrasound leak detection is also possible. After a casting process, the adjustable hose clamping arrangement, particularly the clamping device, is loosened and stored for the next cycle. Additionally, the steel branch (steel bushing) is simply cleaned by removing remaining butyl without the need of any sharp tools. Hence, by using the adjustable hose clamping arrangement, no sharp tools and no cold shrink tubes are needed. Therefore, a better safety of the vacuum-tight sealing and less costs may be providable by the adjustable hose clamping arrangement. Additionally, as the vacuum-tight sealing is visible, the overall process safety may be improved. Further, ergonomics may also be improved because no pulling of the flexible insert (hose) in a cold shrink tube is necessary which is both tricky and hard to the hands and fingers.

[0011] According to this application, the term “fixedly clamping the hose such that a relative movement between the hose and the clamping device is inhibited” may particularly denote that the clamping device may induce a clamping force to the hose which may be such that the hose may not move, particularly relative to the clamping device, during a sealing process.

[0012] The term “clamping device” may denote a device which is movable between a clamping configuration in which the hose is clamped at the clamping device and a releasing configuration in which the hose may be releasable from the clamping device. The clamping device according to embodiments of the present invention may be operatable by a user in a manual operation.

[0013] Furthermore, the term “vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold” may particularly denote that the sealing between an outer surface of the hose and the steel branch is provided such that no air passes from the environment to an inner volume of the vacuum mold through the vacuum-tight sealing. Particularly, the provided vacuum-tight sealing is able to withstand a pressure difference between the ambient pressure outside the mold and surrounding the hose and the pressure inside the vacuum mold. The ambient pressure may particularly be 1013 mbar and the pressure inside the vacuum mold may particularly be 20 mbar. Hence, the vacuum-tight sealing may inhibit air to enter the vacuum mold even under influences of a pressure difference of nearly 1000 mbar.

[0014] The term “mold bracket” may denote the part of the adjustable hose clamping arrangement which is in use mounted to the steel branch of the vacuum mold to form a counterpart for the two connecting devices. The mold bracket is mounted to the steel branch in a first step. Afterwards, the hose is clamped by the clamping device and the clamping device and the mold bracket are coupled by the two connecting devices.

[0015] Further, the term “steel branch of the mold” may be a part of the vacuum mold, which may mounted to the vacuum mold when the two mold halves are fixed together. The steel branch may be releasably fixed to the two mold halves. Further, the steel branch may comprise a tube through which the hose of the flexible insert of the vacuum mold is guidable from the inside of the vacuum mold to the outside of the vacuum mold.

[0016] The term “connecting device” may denote a device which is formed to releasably connect the clamping device to the mold bracket. The two connecting devices may particularly be identical connecting devices. Particularly, the connecting devices may be configured to transmit identical forces, particularly, pulling forces. Further, the connecting device may enable a pulling force from the clamping device on the flexible insert such that a flush mount of an inner flange of the flexible insert towards an inner surface of the vacuum mold, in particular the rigid mold shell may be providable.

[0017] Furthermore, the term “connected to the clamping device” respectively “connected to the mold bracket” may denote a detachable and re-attachable connection. In other words, the two connecting devices may each be re-usable.

[0018] The term “an outer surface of the hose next to a free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch” may denote that the free end of the steel branch and the adjacent part of the hose are positioned in-between the clamping device and the mold branch when the clamping device is in a clamping configuration in which the hose is fixedly clamped and a relative movement between the hose and the clamping device is inhibited. Additionally, the distance between the clamping device and the mold bracket and therefore the length of the two connecting devices is such that a user may access the free end of the steel branch with his hand.

[0019] According to an aspect of embodiments of the present invention, a mold arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold is described. The mold arrangement comprises an above-described adjustable hose clamping arrangement, and a vacuum mold comprising the flexible insert, the hose, and the steel branch, wherein the hose is fixedly clamped by the clamping device such that a relative movement between the hose and the clamping device is inhibited, wherein the mold bracket is mounted to the steel branch of the vacuum mold, wherein the first end of each of the two connecting devices is connected to the clamping device and the second end of each of the two connecting devices being opposite of the first end, is connected to the mold bracket, and wherein the outer surface of the hose next to the free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.

[0020] According to an aspect of embodiments of the present invention, a method of manufacturing an adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, wherein the flexible insert is guidable through the steel branch from an inside of the vacuum mold to an outside of the vacuum mold, is described. In embodiments, the method comprises (a) providing a clamping device configured for fixedly clamping the hose such that a relative movement between the hose and the clamping device is inhibited, (b) providing a mold bracket configured for being mountable to the steel branch of the vacuum mold, (c) providing two connecting devices, (d) connecting a first end of each of the two connecting devices to the clamping device, and (e) connecting a second end of each of the two connecting devices being opposite of the first end, to the mold bracket such that an outer surface of the hose next to a free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.

[0021] According to an aspect of embodiments of the present invention, a method of using an adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold is described. In embodiments, the method of using the adjustable hose clamping arrangement comprises (a) guiding the flexible insert through the steel branch from an inside of the vacuum mold to an outside of the vacuum mold, (b) fixedly clamping the hose by a clamping device such that a relative movement between the hose and the clamping device is inhibited, (c) mounting a mold bracket to the steel branch of the vacuum mold, (d) connecting a first end of each of the two connecting devices to the clamping device, and (e) connecting a second end of each of the two connecting devices being opposite of the first end, to the mold bracket, and (f) vacuum-tight sealing, by using a tacky tape, the hose and the free end of the steel branch.

[0022] According to an exemplary embodiment of the present invention, the clamping device comprises a first clamping element and a second clamping element, wherein the first clamping element and the second clamping element are configured for interacting in a clamping configuration such that the relative movement between the hose and the clamping device is inhibited.

[0023] Hence, the clamping device may comprise the first clamping element and the second clamping element which may be adapted and/or selected to fit a specific hose, particularly a specific outer shape of a hose. Therefore, the adjustable hose clamping arrangement may be used with different hose diameters. Therefore, a flexible clamping device may be providable. Additionally, the hose may be easily inserted in-between the first clamping element and the second clamping element when the first clamping element and the second clamping element are in a non-clamping configuration, namely in a releasing configuration. Thereby, an easy-to-handle system may be providable.

[0024] The term “the first clamping element and the second clamping element are configured for interacting” may denote that the first clamping element and the second clamping element may be formed with corresponding geometries such that the first clamping element and the second clamping element may be formed complementary to each other.

[0025] The relative movement between the hose and the clamping device may be inhibited by inhibiting a relative movement between the first clamping element and the hose and at the same time by inhibiting a relative movement between the second clamping element and the hose.

[0026] According to an exemplary embodiment of the present invention, the first clamping element comprises a first retainer shaped semi-circular and comprising a first diameter, wherein the first diameter is substantially equal to a diameter of the hose, wherein the second clamping element comprises a second retainer shaped semi-circular and comprising a second diameter, wherein the second diameter is substantially equal to the first diameter, and wherein the first retainer and the second retainer are configured for being in contact with the outer surface of the hose when the clamping device is in the clamping configuration.

[0027] Hence, the clamping device may be easy to manufacture and at the same time may provide a reliable clamping of the hose.

[0028] By providing the first diameter of the first retainer and the second diameter of the second retainer being substantially equal may allow the clamping device, when being in the clamping configuration, the first retainer and the second retainer to form a circular opening in-between for holding and clamping the hose.

[0029] By providing the first diameter and the second diameter both substantially equal to the diameter of the hose may allow an easy and reliable clamping of the hose.

[0030] The term “the first retainer and the second retainer are configured for being in contact with the outer surface of the hose” may denote that the respective semi-circular shaped retainer is configured for contacting with the outer surface of the hose. Particularly, an inner contact surface of the respective retainer may have a surface roughness which inhibits a sliding of the hose when the hose is clamped by the clamping device in the clamping configuration.

[0031] According to an exemplary embodiment of the present invention, the first clamping element and the second clamping element are further configured for being distanced to each other such that the hose is releasable from the first clamping element and the second clamping element, when the clamping device is in a releasing configuration.

[0032] Hence, the clamping device may be moveable from the clamping configuration to the releasing configuration by the user in an easy way. Therefore, the clamping device may be easily usable. Particularly, the hose may be releasable from the clamping device in an easy way and in a one hand use.
[0033] The term “releasing configuration” may denote a position at which the first clamping element and the second clamping element are distanced to each other such that the hose may be insertable in-between the first clamping element and the second clamping element. Additionally, the hose may be releasable from the first clamping element and the second clamping element. In an embodiment, the hose may be “laid” in contact with one of the first clamping element, particularly the first retainer, or the second clamping element, particularly the second retainer, and the other one of the first clamping element and the second clamping element is brought in contact with the other clamping element.
[0034] According to an exemplary embodiment of the present invention, the first clamping element and/or the second clamping element comprises a mechanical stop configured for preventing a crushing of the first clamping element and the second clamping element when the clamping device is moved to the clamping configuration.
[0035] Therefore, a durable clamping device may be providable. Additionally, the hose may be protected and damages of the hose may be avoided. Hence, a reliable sealing may be providable.
[0036] The term “mechanical stop” may denote a feature with absorbs potential forces acting on the first clamping element and the second clamping element when the first clamping element and the second clamping element come in contact. Particularly, the mechanical stop may be a built-in physical stop preventing a crushing of the first clamping element and the second clamping element. The mechanical stop may particularly be an elastic capping.
[0037] According to an exemplary embodiment of the present invention, the first clamping element further comprises a fastening portion coupled to the first retainer and comprising two through-holes, wherein the fastening portion is configured for connecting the first end of each of the two connecting devices to a respective one of the two through-holes.
[0038] Therefore, a convenient clamping device may be providable. The clamping device additionally allows clamping of the hose or adjusting of the clamping of the hose when the clamping device has already been mounted to the mold bracket by the two connecting devices.
[0039] The term “connecting the first end of each of the two connecting devices to a respective one of the two through-holes” may denote that the connecting device is inserted through the though-hole such that the head lies against the surface of the fastening portion being turned away from the mold bracket when the clamping device is mounted to the mold bracket. Further, the fastening portion is configured for connecting the first end of each of the two connecting devices to a respective one of the two through holes such that, when the clamping device is in the clamping configuration, the two connecting devices each extend parallel to the fixedly clamped hose.
[0040] According to an exemplary embodiment of the invention, the clamping device further comprises a handling portion configured for handling the clamping device by a user and configured for adjusting a clamping force of the first clamping element and the second clamping element, and a first arm connecting the handling portion and the first clamping element, a second arm connecting the handling portion and the second clamping element, wherein the first arm and the second arm are connected to the handling portion by a hinge configured for moving the first clamping element and the second clamping element between the clamping configuration and the releasing configuration.
[0041] Hence, the clamping device may be easily operatable in a one hand use by a user (e.g., with one hand only).
[0042] The term “adjusting the clamping force” may denote that different clamping forces and thereby retaining forces may be adjustable by the user dependent on the intended use, particularly dependent on properties of the hose to be clamped. In embodiments, the material of the different

hoses may withstand different clamping respectively retaining forces. Additionally, the clamping force may be dependent on the pressure difference applied to the vacuum mold.
[0043] According to an exemplary embodiment of the invention, the first arm and the second arm are each formed in a C-shape. Thereby, the first arm and the second arm may be formed identical. Particularly, the first arm and the second arm may each be coupled to the handling portion at a common hinge. Further, the first clamping element may be exchangeable coupled to the first arm at a first fastening point. Additionally, the second clamping element may be coupled to the second arm at a second fastening point.
[0044] The term “the hinge is configured for moving the first clamping element and the second clamping element between the clamping configuration and the releasing configuration” may denote that the hinge may allow an angular movement of the first arm relative to the second arm. Thereby, the first clamping element and the second clamping element may be moved between the clamping configuration in which the first clamping element and the second clamping element are in contact with each other, and the releasing configuration in which the first clamping element and the second clamping element are distanced to each other.
[0045] According to an exemplary embodiment of the present invention, the first clamping element is exchangeable mounted to the first arm, and the second clamping element is exchangeable mounted to the second arm.
[0046] Therefore, a flexible clamping device may be providable which is usable for different diameters of the hose and hence for different vacuum molds.
[0047] The term “exchangeable mounted” may denote that the clamping element may be dismounted, and another different clamping element may be mounted to the arm. Particularly, a clamping element may be mounted comprising a different diameter. Thereby, if a hose having a larger diameter should be clamped in the clamping configuration, the first clamping element and the second clamping element may be displaced by different clamping elements having a first retainer and a second retainer being correspondingly semi-circular shaped and having a larger diameter.
[0048] According to an exemplary embodiment of the present invention, each of the two connecting devices comprises a screw, particularly a finger screw, comprising a head, particularly a knurled head, at the first end, a compression spring arranged at the screw, and configured for exerting a pulling force on the hose, when the hose is fixedly clamped in the clamping configuration, and a guide sleeve extending over substantially an entire axial extension of the screw, wherein the screw and the compression spring are arranged inside the guide sleeve, and wherein the first end of each of the guide sleeves is connectable to the clamping device and the second end of each of the guide sleeves is connectable to the mold bracket.
[0049] Hence, an easy and reliable connecting device may be providable which enables at the same time connecting the clamping device and the mold bracket and providing a pulling force on the hose and thereby on a flange of the flexible insert. Hence, a flush mounting of the flange of the flexible insert against an inner surface of the mold may be providable. Thereby, the vacuum-tight sealing may be improved.
[0050] Providing the screw with a knurled head at the first end may provide an ergonomic possibility to mount the screw.
[0051] The term “exerting a pulling force on the hose” may denote that the compression spring is arranged at the screw such that, when the hose is fixedly clamped by the clamping device, the pulling fore is exerted on the hose. Thereby, the flange of the flexible insert is mounted flush against an inner wall of the mold shell.
[0052] Further, the term “guide sleeve” may denote a guide tap or a guide bushing. The guide sleeve may be a ridged hollow body having an external thread on its outer surface. Particularly, an external thread on at least a part of the outer surface which is intended to be mounted to the mold bracket.

[0053] By providing each of the two compression springs on two opposing sides of the first retainer may provide an equal and uniform force transmission on the hose.
[0054] The term "the first end is connected to the clamping device" may denote that the head of the screw lies against a surface of the clamping device turned away from the mold bracket, when the clamping device and the mold bracket are mounted to each other by the two connecting devices.
[0055] According to an exemplary embodiment of the present invention, the mold bracket comprises two threaded guide holes, wherein each of the two threaded guide holes is configured for mounting, particularly screw mounting, of a respective one of the two connecting devices, wherein the second end of the guide sleeve is configured for fitting in a respective one of the two threaded guide holes.
[0056] Hence, a reliable detachably fixing of the connecting device and the mold bracket may be provided.
[0057] The term "threaded guide hole" may denote a blind hole comprising an internal thread along at least a part of its axial extension.
[0058] Furthermore, the term "the second end of the guide sleeve is configured for fitting in a respective one of the two threaded guide holes" may denote that the outer diameter of the guide sleeve corresponds substantially to the inner diameter of the threaded guide hole. Additionally, the second end of the guide sleeve may comprise an external thread corresponding to an internal thread of the threaded guide hole.
[0059] The term "screw mounting" may denote a connection of an external thread with a corresponding internal thread. Particularly, the screw mounting may be detachably fixing the respective connecting device inside the respective threaded guide hole.
[0060] According to an exemplary embodiment of the invention, the first end of each of the two guide sleeves is fixed in one of the two through-holes in the fastening portion.
[0061] Therefore, a reliable connection between the mold bracket and the clamping device may be providable. At the same time an exchangeable connecting device may be usable dependent on for example the intended pulling force or retaining force.
[0062] The term "the first end of the guide sleeve is fixed in the through-hole in the fastening portion" may denote that the first end of the guide sleeve is guided through the through-hole and the head of the screw lies against the surface of the fastening portion being turned away from the mold bracket when the clamping device and the mold bracket are mounted to each other by the connecting devices.
[0063] According to an exemplary embodiment of the present invention, the mold bracket comprises two halves which are connectable to each other, in particular by a screw connection, wherein each of the two halves comprises a semi-circular recess comprising a recess diameter, wherein the recess diameter is substantially equal to an outer diameter of the steel branch of the mold, and wherein the two semi-circular recesses are configured for enclosing the steel branch of the mold, when the mold bracket is mounted to the steel branch, wherein in particular one of the two halves comprises two threaded guide holes configured for mounting, particularly screw mounting, of a respective one of the connecting devices.
[0064] Hence, the mold bracket may be easily mounted and disassembled from the mold. Thereby, a re-usable mold bracket may be providable which saves costs.
[0065] Particularly, the screw connection may comprise two screws mounted to opposing sides of the semi-circular recess.
[0066] By providing the two threaded guide holes in one of the two halves may allow to solely exchange one half if, e.g., the threaded guide hole is damaged, and a fastening of the guide sleeve may no longer be possible. Hence, an easy and cost-efficient maintenance of the entire adjustable hose clamping arrangement may be possible.

Description

BRIEF DESCRIPTION

[0067] Some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members, wherein:

[0068] FIG. **1** shows an adjustable hose clamping arrangement with a hose and a steel branch according to an exemplary embodiment;

[0069] FIG. **2** shows a clamping device according to an exemplary embodiment;

[0070] FIG. **3** shows in a side view an adjustable hose clamping arrangement according to an exemplary embodiment of the invention;

[0071] FIG. **4** shows in a perspective view an adjustable hose clamping arrangement according to an exemplary embodiment; and

[0072] FIG. **5** shows a side view of a mold arrangement according to an exemplary embodiment of the invention.

DETAILED DESCRIPTION

[0073] FIG. **1** shows an adjustable hose clamping arrangement **100** with a hose **182** and a steel branch **191** according to an exemplary embodiment. The adjustable hose clamping arrangement **100** may be used for vacuum-tight sealing of the hose **182** of a flexible insert **584** (shown in FIG. **5**) to a free end **192** of a steel branch **191** of a vacuum mold mold **593** (shown in a partial view in FIG. **5**), wherein the flexible insert **584** is guidable through the steel branch **191** from an inside of the vacuum mold **593** to an outside of the vacuum mold **593**. The adjustable hose clamping arrangement **100** comprises a clamping device **110** fixedly clamping the hose **182** such that a relative movement between the hose **182** and the clamping device **110** is inhibited, a mold bracket **140** mounted to the steel branch **191** of the vacuum mold **593**, a first connecting device **120** and a second connecting device **120** being equal to the first connecting device **120**. In FIG. **1** solely the head **121** of the second connecting device **120** is shown. A first end **151** of the first connecting device **120** is connected to the clamping device **110** and a second end **152** of the first connecting device **120** being opposite of the first end **151**, is connected to the mold bracket **140** such that an outer surface **186** of the hose **182** next to a free end of the steel branch **191** is accessible for the vacuum-tight sealing between the hose **182** and the free end of the steel branch **191**.

[0074] FIG. **2** shows a clamping device **110** according to an exemplary embodiment. The clamping device **110** comprises a first clamping element **211** and a second clamping element **212**. In FIG. **2**, the clamping device **110** is shown in a clamping configuration without the hose **182**. The first clamping element **211** and the second clamping element **212** are configured for interacting in the illustrated clamping configuration such that the relative movement between the hose **182** (not illustrated in FIG. **2**) and the first clamping element **211** as well as the second clamping element **212** is inhibited.

[0075] The clamping device **110** further comprises a handling portion **273** configured for handling the clamping device **110** by a user and configured for adjusting a clamping force of the first clamping element **211** and the second clamping element **212**. The handling portion **273** and the first clamping element **211** are connected by a first arm **271**. Furthermore, the second clamping element **212** and the handling portion **273** are connected by a second arm **272**. The first arm **271** and the second arm **272** are formed in a C-shape. The first arm **271** and the second arm **272** are coupled to the handling portion **273** by a common hinge **276** which allows a movement of the first arm **271** and the second arm **272** between the clamping configuration (shown in FIG. **2**) and a releasing configuration in which the first clamping element **211** and the second clamping element **212** are distanced to each other such that the hose **182** may be released from the first clamping element **211** and the second clamping element **212**.

[0076] The first clamping element **211** is coupled to the first arm **271** at a first fastening point **275** which is configured for allowing an exchangeable coupling of the first clamping element **211** to the first arm **271**. Particularly, the first fastening point **275** comprises a screw connection between the first arm **271** and the first clamping element **211**. Further, the second clamping element **212** is coupled to the second arm **272** at a second fastening point **274** which is configured for allowing an exchangeable coupling of the second clamping element **212** to the second arm **272**. Particularly, the second fastening point **274** comprises a screw connection between the second arm **272** and the second clamping element **212**.

[0077] The second clamping element **212** comprises a second retainer **214** which is shaped semi-circular and comprises a second diameter **478** (illustrated in FIG. **4**). The second diameter **478** is substantially equal to the outer diameter **185** of the hose **182**. The second retainer **214** comprises a semi-circular shaped second inner contact surface **219** which is configured for being in contact with the outer surface **186** of the hose **182** when the clamping device **110** is in the clamping configuration. The second inner contact surface **219** may be roughened to enhance an ability of inhibiting a relative movement between the second retainer **214** respectively the second inner contact surface **219** and the outer surface **186** of the hose **182**.

[0078] The first clamping element **211** comprises a first retainer **213** which is semi-circular shaped and comprises a first diameter **477** (illustrated in FIG. **4**) corresponding to the outer diameter **185** of the hose **182**. The first diameter **477** and the second diameter **478** are substantially equal. The first retainer **213** comprises a semi-circular shaped first inner contact surface **217** which is configured for being in contact with the outer surface **186** of the hose **182** when the clamping device **110** is in the clamping configuration. The first inner contact surface **217** and the second inner contact surface **219** may be identical for allowing a more even force transmission and clamping of the hose **182**.

[0079] Furthermore, the first clamping element **211** comprises a fastening portion **218** coupled to the first retainer **215**. As shown in FIG. **2**, the fastening portion **218** and the first retainer **213** may be formed in one piece. The fastening portion **218** comprises two through-holes **279**. As illustrated in FIG. **2**, the two through-holes **279** may be formed on two opposing sides of the first retainer **213**.

[0080] The first clamping element **211** comprises a mechanical stop **216** being formed as a part of the first retainer **213** at each of the two ends of the semi-circular shaped first retainer **215**. Hence, in total the first retainer **213** comprises two mechanical stops **216**. Additionally, the second clamping element **212** comprises a mechanical stop **216** being formed as a part of the second retainer **214** at each of the two ends of the semi-circular shaped second retainer **214**. Hence, in total the second retainer **214** comprises two mechanical stops **216**. The mechanical stops **216** of the first clamping element **211** and the mechanical stops **216** of the second clamping element **212** comprise mating geometries. Thereby, a crushing of the first retainer **213** and the second retainer **214** may be inhibited.

[0081] FIG. **3** shows in a side view an adjustable hose clamping arrangement **100** according to an exemplary embodiment of the invention. The adjustable hose clamping arrangement **100** comprises a clamping device **110**, two connecting devices **120** and the mold bracket **140**. In FIG. **3** solely one of the two connecting devices **120** is shown. As may be seen in FIG. **3**, the mechanical stop **216** formed as a part of the first retainer **213** and the mechanical stop **216** formed as a part of the second retainer **214** comprise a mating geometry for inhibiting a crushing of the first retainer **213** and the second retainer **214**.

[0082] The mold bracket **140** comprises a first half **341** and a second half **342**, both illustrated in a sectional view in FIG. **3**. The first half **341** and the second half **342** are mounted to each other by two screws **344**, wherein in FIG. **3** solely one screw **344** is shown. The first half **341** comprises two threaded guide holes **343**, wherein in FIG. **3** solely one of the two threaded guide holes **343** is illustrated.

[0083] The connecting device **120** comprises a knurled head **121** at a first end **151** and is inserted through the through-hole **279** (shown in FIG. **2**) of the fastening portion **218** of the first clamping element **211**. Thereby, an outer surface of a guide sleeve **324** is at the first end **151** in contact with the inner surface of the through-hole **279**. At the second end **152**, the guide sleeve **324** is fitted into and mounted to the threaded guide hole **343** of the first half **341** of the mold bracket **140**. Additionally, a screw **322** and a compression spring **323** are arranged inside the guide sleeve **324**, whereby the compression spring **323** is arranged around the screw **322** and extends over a part of the axial extension of the screw **322**. The compression spring **323** may provide a pulling force on the hose **182** and therefore on the flange **181** of the flexible insert **584** of the vacuum mold **593** when the hose **182** is fixedly clamped in the clamping configuration. The screw **322** mounts the connecting device **120** to the mold bracket **140**.

[0084] FIG. **4** shows in a perspective view an adjustable hose clamping arrangement **100** according to an exemplary embodiment. The adjustable hose clamping arrangement **100** comprises a clamping device **110** which is described in more detail with regard to FIG. **2**, two connecting devices **120** which are described in more detail with regard to FIG. **3**, and the mold bracket **140**.

[0085] The first retainer **213** comprises a first diameter **477** and the second retainer **214** comprises a second diameter **478**. As may be seen in FIG. **4**, the first diameter **477** and the second diameter **478** are substantially equal. The first diameter **477** and the second diameter **478** are chosen dependent on the outer diameter **185** of the hose **182** to be clamped.

[0086] The mold bracket **140** comprises the first half **341** and the second half **342** which are connected to each other by two screws **431**. Each screw **431** is screwed in an internal thread of a fixation hole **432** in the second half **342**.

[0087] The first half **341** comprises a first semi-circular recess **445** and the second half **342** comprises a second semi-circular recess **446**. The first recess **445** and the second recess **446** both comprise a recess diameter **447** being substantially equal to an outer diameter **594** of the steel branch **191** (both illustrated in FIG. **5**). The two semi-circular recesses **445** and **446** are configured for enclosing the steel branch **191** of the mold **593**, when the mold bracket **140** is mounted to the steel branch **191**.

[0088] The second half **342** comprises the two threaded guide holes **343** (solely one is illustrated in FIG. **4**) to which a respective connecting device **120** is screw mounted.

[0089] FIG. **5** shows a side view of a vacuum mold arrangement **500** according to an exemplary embodiment of the invention. The mold arrangement **500** comprises the adjustable hose clamping arrangement **100** and a vacuum mold **593**. The mold bracket **140** is mounted to the steel branch **191** which provides a connection from the mold cavity shown on the right side of the vacuum mold **593** to the environment shown on the left side of the vacuum mold **593** in FIG. **5**.

[0090] The clamping device **110** is mounted by the two connecting devices **120** to the mold bracket **140** which is mounted to the steel branch **191** of the mold **593**. The flexible insert **584** is fixed to the flange **191** such that after vacuum-tight sealing of the hose **182** to the steel branch **191**, a molding material may be inserted in the vacuum mold **593** via the hose **182**.

[0091] At the free end **192** of the steel branch **191**, the hose **182** may be vacuum-tight sealed to the steel branch **191** in a sealing area **583** illustrated by the dotted line. As may be seen in FIG. **5**, the sealing area **583** may be conveniently accessed. By choosing a different length of the connecting devices **120**, the distance between the clamping device **110** and the free end **192** of the steel branch **191** may be adapted.

[0092] Although the present invention has been disclosed in the form of embodiments and variations thereon, it will be understood that numerous additional modifications and variations could be made thereto without departing from the scope of the invention.

[0093] For the sake of clarity, it is to be understood that the use of “a” or “an” throughout this application does not exclude a plurality, and “comprising” does not exclude other steps or elements.

Claims

1. An adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, wherein the flexible insert is guidable through the steel branch from an inside of the vacuum mold to an outside of the vacuum mold, wherein the adjustable hose clamping arrangement comprises: a clamping device configured for fixedly clamping the hose such that a relative movement between the hose and the clamping device is inhibited; a mold bracket configured for being mountable to the steel branch of the vacuum mold; and two connecting devices, wherein a first end of each of the two connecting devices is connected to the clamping device and a second end of each of the two connecting devices being opposite of the first end, is connected to the mold bracket such that an outer surface of the hose next to the free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.
2. The adjustable hose clamping arrangement according to claim 1, wherein the clamping device comprises a first clamping element; and a second clamping element, wherein the first clamping element and the second clamping element are configured for interacting in a clamping configuration such that the relative movement between the hose and the clamping device is inhibited.
3. The adjustable hose clamping arrangement according to claim 2, wherein the first clamping element comprises a first retainer shaped semi-circular and comprising a first diameter, wherein the first diameter is substantially equal to an outer diameter of the hose, wherein the second clamping element comprises a second retainer shaped semi-circular and comprising a second diameter, wherein the second diameter is substantially equal to the first diameter; and wherein the first retainer and the second retainer are configured for being in contact with the outer surface of the hose when the clamping device is in the clamping configuration.
4. The adjustable hose clamping arrangement according to claim 2, wherein the first clamping element and the second clamping element are further configured for being distanced to each other such that the hose is releasable from the first clamping element and the second clamping element, when the clamping device is in a releasing configuration.
5. The adjustable hose clamping arrangement according to claim 2, wherein the first clamping element and/or the second clamping element comprises a mechanical stop configured for preventing a crushing of the first clamping element and the second clamping element when the clamping device is moved into the clamping configuration.
6. The adjustable hose clamping arrangement according to claim 3, wherein the first clamping element further comprises: a fastening portion coupled to the first retainer and comprising two through-holes, wherein the fastening portion is configured for connecting the first end of each of the two connecting devices to a respective one of the two through-holes.
7. The adjustable hose clamping arrangement according to claim 2, wherein the clamping device further comprises: a handling portion configured for handling the clamping device by a user and configured for adjusting a clamping force of the first clamping element and the second clamping element, and a first arm connecting the handling portion and the first clamping element, a second arm connecting the handling portion and the second clamping element, wherein the first arm and the second arm are connected to the handling portion by a hinge, wherein the hinge is configured for moving the first clamping element and the second clamping element between the clamping configuration and the releasing configuration.
8. The adjustable hose clamping arrangement according to claim 7, wherein the first clamping element is exchangeable mounted to the first arm, and wherein the second clamping element is exchangeable mounted to the second arm.
9. The adjustable hose clamping arrangement according to claim 1, wherein each of the two connecting devices comprises: a screw, comprising a head at the first end; a compression spring arranged at the screw, and configured for exerting a pulling force on the hose, when the hose is

fixedly clamped in the clamping configuration; and a guide sleeve extending over substantially an entire axial extension of the screw, wherein the screw and the compression spring are arranged inside the guide sleeve, wherein the first end of each of the guide sleeves is connected to the clamping device and the second end of each of the guide sleeves is connected to the mold bracket.

10. The adjustable hose clamping arrangement according to claim 9, wherein the mold bracket comprises two threaded guide holes, wherein each of the two threaded guide holes is configured for mounting of a respective one of the two connecting devices, wherein the second end of the guide sleeve is configured for fitting in a respective one of the two threaded guide holes.

11. The adjustable hose clamping arrangement according to claim 10, wherein the first end of each of the two guide sleeves is fixed in one of the two through-holes in the fastening portion.

12. The adjustable hose clamping arrangement according to claim 1, wherein the mold bracket comprises two halves which are connectable to each other, wherein each of the two halves comprises a semi-circular recess comprising a recess diameter, wherein the recess diameter is substantially equal to an outer diameter of the steel branch of the vacuum mold, and wherein the two semi-circular recesses are configured for enclosing the steel branch of the vacuum mold, when the mold bracket is mounted to the steel branch, wherein one of the two halves comprises two threaded guide holes configured for mounting of a respective one of the connecting devices.

13. A vacuum mold arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, the vacuum mold arrangement comprising: an adjustable hose clamping arrangement according to claim 1; and a vacuum mold comprising the flexible insert, the hose, and the steel branch, wherein the hose is fixedly clamped by the clamping device such that a relative movement between the hose and the clamping device is inhibited, wherein the mold bracket is mounted to the steel branch of the vacuum mold, wherein the first end of each of the two connecting devices is connected to the clamping device and the second end of each of the two connecting devices being opposite of the first end, is connected to the mold bracket, and wherein the outer surface of the hose next to the free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.

14. A method of manufacturing an adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, wherein the flexible insert is guidable through the steel branch from an inside of the vacuum mold to an outside of the vacuum mold, wherein the method comprises: providing a clamping device configured for fixedly clamping the hose such that a relative movement between the hose and the clamping device is inhibited; providing a mold bracket configured for being mountable to the steel branch of the vacuum mold; providing two connecting devices; and connecting a first end of each of the two connecting devices to the clamping device, and connecting a second end of each of the two connecting devices being opposite of the first end, to the mold bracket such that an outer surface of the hose next to a free end of the steel branch is accessible for the vacuum-tight sealing between the hose and the free end of the steel branch.

15. A method of using an adjustable hose clamping arrangement for vacuum-tight sealing of a hose of a flexible insert to a free end of a steel branch of a vacuum mold, the method of using the adjustable hose clamping arrangement comprises: guiding the flexible insert through the steel branch from an inside of the vacuum mold to an outside of the vacuum mold; fixedly clamping the hose by a clamping device such that a relative movement between the hose and the clamping device is inhibited; mounting a mold bracket to the steel branch of the vacuum mold; and connecting a first end of each of the two connecting devices to the clamping device; and connecting a second end of each of the two connecting devices being opposite of the first end, to the mold bracket, vacuum-tight sealing, by using a tacky tape, the hose and the free end of the steel branch.